Sampling design and weighting strategies in the Values in Crisis data for Austria 2020

1. Introduction

The following document describes the sampling design and weighting strategy for the Austrian Values in Crisis (VIC Austria) sample. First the target population will be characterized. Next, the applied sampling strategy is explained in detail to set the stage for the discussion of coverage and sampling errors and the corrections by post-stratification weights.

2. What population does VIC Austria represent?

The survey was conducted by the Marketagent.com online reSEARCH GmbH (Marketagent). The sample was drawn from a consumer online panel with 128,500 participants in Austria. The online panel integrates persons who are at least 14 years or older. All educational groups, professions and Austrian regions are well integrated in the panel although people in older age groups and immigrants are clearly underrepresented. The participants speak (one of) the official language(s) of the country (regardless of nationality and citizenship) and do not live either abroad or in institutions such as prisons and hospitals during the entire fieldwork period.

Table 1: Comparison Sampling frame and target population

	Characteristic	Marketagent Panel[1]	Statistics Austria[2]
Female	14-19 years	2.8	3.3
	20-29 years	15.7	7.2
	30-39 years	16.8	7.7
	40-49 years	9.9	7.9
	50-59 years	7.1	9.0
	60+ years	3.8	16.0
	Sum	56.2	51.2
Male	14-19 years	2.6	3.5
	20-29 years	12.0	7.6
	30-39 years	10.6	7.9
	40-49 years	7.6	7.9
	50-59 years	6.0	9.0
	60+ years	5.1	12.9
	Sum	43.8	48.8
Marital status	Single	58.3	35.2
	Married	30.6	48.3
	Divorced	9.9	7.2
	Widowed	1.2	9.3
Household size	Single person	20.3	37.1
	Two persons	33.2	30.3
	Three and more	46.5	32.6
Children per household	None	56.5	77.2
	One	35.2	10.9
	Two or more	8.3	11.9

Note: [1] Marketagent 2020; [2] Statistics Austria relies on different databases: Gender/Age (Statistics Austria 2020a); Marital status (Statistics Austria 2002b); Household size (Statistics Austria 2020c); Children in Household (Statistics Austria 2020d)

Table 1 illustrates major deviations of the Online-Panel compared to Austrian statistics. Compared to the target population the Marketagent panel has more women and people between 20 and 39 years. Additionally samples drawn from the panel recruiting people 60 years and more will rely on a smaller pool than younger age groups. Similar deviations from the target population appear with regard to marital status, household size and number of children in the household. Married and widowed people are underrepresented as well as people in single person households and households without children. Obviously these deviations relate to the age distortion as older people are more likely to be widowed and to live in a single person household without children.

3. The basic principle of the VIC Austria sampling design?

The VIC Austria is part of a cross-national survey and as most modern population surveys across multiple countries, the different sampling strategies are applied in order to account for country characteristics (Kish 1994). To adjust sampling probabilities weighting based on the full population coverage needs to be implemented.

Table 2: Comparison unweighted sample and target population

	Characteristic	Unweighted Sample	Statistics Austria
Female[1]	14-19 years	3.8	3.3
	20-29 years	7.8	7.2
	30-39 years	8.1	7.7
	40-49 years	8.7	7.9
	50-59 years	9.3	9.0
	60-69 years	7.1	6.7
	70+ years	5.9	9.3
	Sum	50.7	51.2
Male[1]	14-19 years	3.7	3.5
	20-29 years	8.2	7.6
	30-39 years	7.9	7.9
	40-49 years	8.6	7.9
	50-59 years	9.4	9.0
	60-69 years	6.4	6.2
	70+ years	5.0	6.7
	Sum	49.3	48.8
Region[2]	Vorarlberg	4.4	4.5
	Tyrol	8.6	8.5
	Salzburg	6.2	6.3
	Styria	14.3	14.0
	Carinthia	6.5	6.3
	Upper Austria	16.9	16.7
	Lower Austria	18.9	18.9
	Vienna	20.5	21.4
	Burgenland	3.7	3.3
Education[3]	Compulsory education	17.1	25.7
	Apprenticeship	38.7	31.8
	Vocational middle school	13.1	14.0
	General secondary school	7.3	6.5
	Vocational secondary school	12.2	10.3
	University	10.7	11.6

Note: [1] Source: Statistics Austria 2020a; [2] Source: Statistics Austria 2020e; [3] Source: Statistics Austria 2020f.

In order to address these distortions in the online panel, a stratified quota sampling strategy was applied for gender, age, region and education. The distribution of the unweighted final sample is presented in Table 2. As a result of certain quotas, the sample represents the target population already rather well. Still the sampling probabilities for the different groups vary due to the mismatch between sampling frame and target population. This will especially impose a bias for the age group 70+.

To correct for deviations between sample and target population, survey sample weights were calculated by iterative proportional fitting aligning the data to population margins. The reference population margins are based on a three way table of gender, age and education (see Table 3) and on region separately (see Table 2). The STATA package ipfweight by Michael Bergmann (2011) was used to estimate the weights. The calculated weight variable has a standard deviation of 0.667, a minimum of 0.272 and a maximum of 5.325. From the 2003 cases 41 exceed a weight of 3.0 and 18 cases have a weight lower than 0.3. The 41 cases above 3.0 scale up to 192.8 cases and the 18 cases with weight smaller than 0.3 account for 5.2 cases in the weighted sample. Hence, no adjustment was made to the weights to account for outliers (see Little et al. 1997).

Table 3 compares the weighted data in a three-way table of gender, age and education against the unweighted data. The lower share of older people with lower education is corrected by the weighting (see column "Compulsory school" in Table 3). The older people with lower education levels have the biggest weights and interpretations in the data analyses need to reflect on that. The deviations on the other groups are minor and are less problematic.

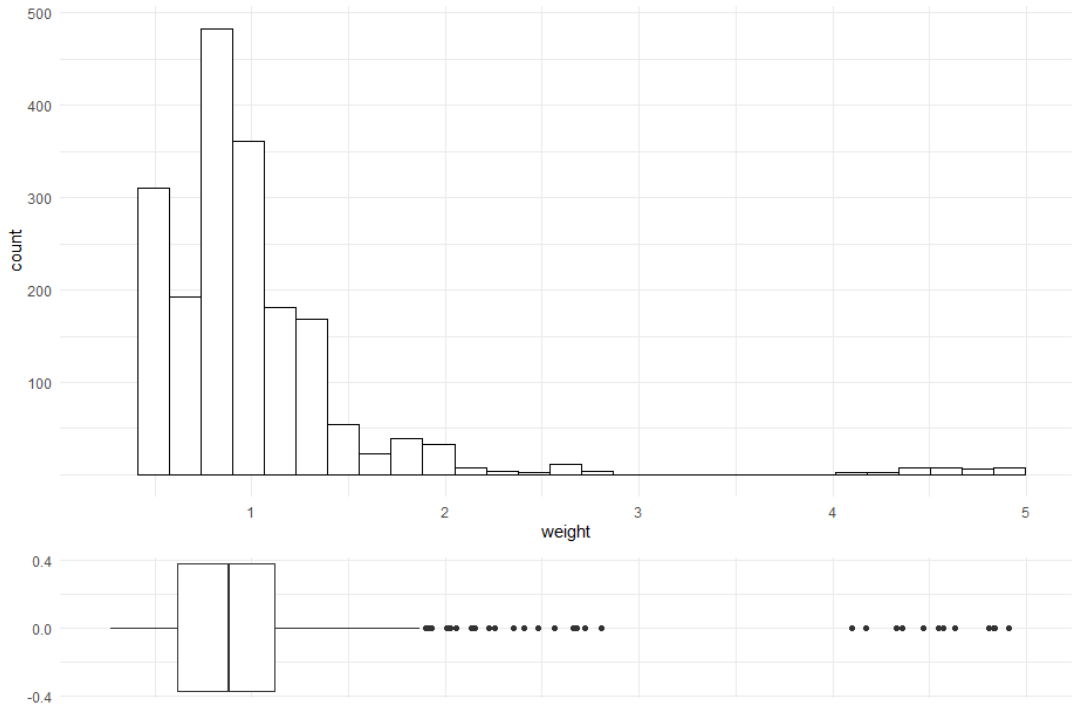

Figure 1: Distribution of the weights

Table 3: Overview weighting by gender, age and education (distributions of weighted and unweighted data)

		Highest completed training (education)											
		Compulsory school		**Apprenticeship**		**Vocational middle school**		**General or vocational secondary school**		**University**		***Total***	
		weighted	*unweighted*	*weighted*	*unweighted*	*weighted*	*unweighted*	*weighted*	*unweighted*	*weighted*	*unweighted*	*weighted*	*unweighted*
Female	**14-29 years**	3.30	3.20	1.82	4.19	1.02	0.70	2.80	2.90	1.30	0.70	*10.24*	*11.69*
	30-39 years	1.30	1.45	1.83	4.04	1.06	1.00	1.56	0.90	2.01	0.75	*7.76*	*8.14*
	40-49 years	1.64	2.25	2.34	2.75	1.42	1.40	1.35	1.10	1.53	1.25	*8.27*	*8.75*
	50-59 years	2.15	1.15	2.59	2.95	1.91	2.05	1.10	1.95	1.25	1.20	*9.00*	*9.30*
	60-69 years	7.02	1.45	1.96	3.25	1.30	2.25	0.48	0.35	0.71	0.30	*15.97*	*13.05*
	70+ years			2.05	1.45	1.53	1.15	0.57	1.90	0.35	0.95		
	Sum	*15.41*	*9.50*	*12.59*	*18.63*	*8.23*	*8.55*	*7.86*	*9.10*	*7.14*	*5.15*	*51.23*	*50.93*
Male	**14-29 years**	4.05	3.30	2.84	3.15	0.83	0.70	2.36	4.04	0.79	0.60	*10.87*	*11.79*
	30-39 years	1.34	1.60	2.87	3.25	0.85	0.70	1.32	1.25	1.56	1.20	*7.93*	*8.00*
	40-49 years	1.16	1.30	3.51	4.29	1.00	0.75	1.18	1.10	1.36	1.10	*8.21*	*8.54*
	50-59 years	1.18	0.85	4.21	4.79	1.28	0.95	1.16	1.85	1.17	1.05	*9.00*	*9.49*
	60-69 years	2.61	0.60	2.93	3.30	0.83	1.10	0.62	1.10	0.76	0.50	*12.76*	*11.35*
	70+ years			2.83	1.50	0.96	0.45	0.59	1.45	0.63	1.35		
	Sum	*10.34*	*7.65*	*19.18*	*20.28*	*5.76*	*4.65*	*7.22*	*10.79*	*6.27*	*5.80*	*48.77*	*49.17*
Total		*25.74*	*17.15*	*31.77*	*38.91*	*13.99*	*13.20*	*15.08*	*19.89*	*13.42*	*10.95*	100.0	100.0

Note: Data basis for weighting: Statistics Austria - STATcube - Coordinated Labour Force Statistics - Persons - Time Series from 2011 (QSW) - Year 2017.

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Appendix STATA Code

```
* ==============================================================================
* PROJECT:          VALUES IN A CRISIS
* SOURCE:           SPSS_Uni Salzburg_Werte in der Krise_Welle 1_Mai 2020.sav
* OBJECTIVE:    WEIGHTS
* DATE:             17.06.2020
* ==============================================================================

clear all
set more off
set maxvar 32767
cd "PATH"
use "DATA"

replace Q6=5 if RQ6A12=="Bhs"                    // Eindeutige offene Angaben zuordnen
replace Q6=9 if RQ6A12=="Mag,rer.nat." |RQ6A12=="Magistra"

generate help_g=Q1*100 if Q1!=3
generate help_a=Q2*10
replace help_a=. if Q3A1==2006
generate help_b=Q6 if Q6<12
replace help_b=5 if RQ6A12=="Kollege" |RQ6A12=="Lehre mit Matura"
replace help_b=6 if RQ6A12=="Akademie"      // Bildungsabschlüsse mit klarer Zuordnung zu den Gewichtungskategorien aber nicht zu den abgefragten Variablen
recode help_b (6 7 8 9 10 11 = 6)
generate help=help_g+help_a+help_b

replace help=help-1 if Q6==5 | RQ6A12=="Kollege" |RQ6A12=="Lehre mit Matura"
replace help=help-10 if Q2==3

replace help=171 if help==181 // Zusammenlegen von Ü70 und 60-69 mit Pflichtschule (Männer)
```

```
replace help=271 if help==281 // Zusammenlegen von Ü70 und 60-69 mit Pflichtschule (Frauen)

ipfweight help Q4, gen(weight) ///
val(	4.05 2.84 0.83 2.36 0.79 ///
		1.34 2.87 0.85 1.32 1.56 ///
		1.16 3.51 1.00 1.18 1.36 ///
		1.18 4.21 1.28 1.16 1.17 ///
		2.61 2.93 0.83 0.62 0.76 ///
			2.83 0.96 0.59 0.63 ///
		3.30 1.82 1.02 2.80 1.30 ///
		1.30 1.83 1.06 1.56 2.01 ///
		1.64 2.34 1.42 1.35 1.53 ///
		2.15 2.59 1.91 1.10 1.25 ///
		7.02 1.96 1.30 0.48 0.71 ///
			2.05 1.53 0.57 0.35 ///
4.5 8.5 6.3 14.0 6.3 16.7 18.9 21.4 3.3 ///  /* Bundesland */
) maxit(100) tol(.1)

rename help* w1_help*
```